**5** **(a)** A capacitor of capacitance $C_1$ is connected in series with a second capacitor of capacitance $C_2$.

Show that the combined capacitance $C$ of the two capacitors is given by

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}.$$

[2]

**(b)** Three identical capacitors, each of capacitance $C$, are connected in a network as shown in Fig. 5.1.

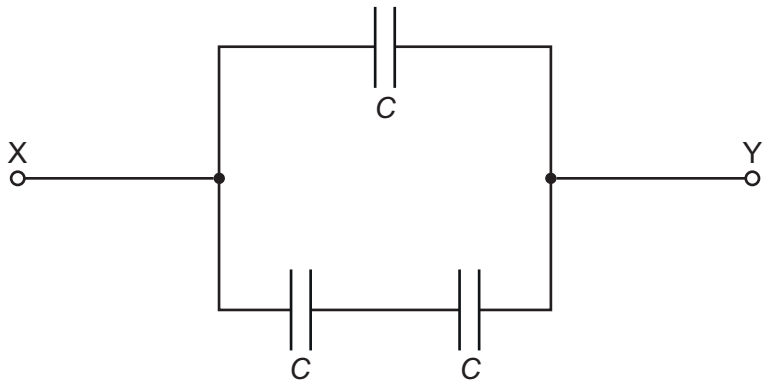



**Fig. 5.1**

The variation of the charge $Q$ with the potential difference (p.d.) $V$ between the terminals X and Y is shown in Fig. 5.2.

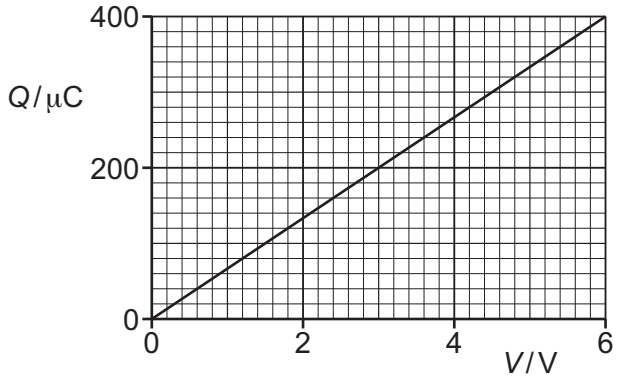



**Fig. 5.2**

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Show that $C$ is equal to 44 μF.

[3]

**(c)** The capacitor network in Fig. 5.1 is charged and then connected to a resistor of resistance 54 kΩ. The capacitor network discharges through the resistor.

**(i)** Determine the time constant $\tau$ of the circuit. Give a unit with your answer.

$\tau$ = ..... unit ..... [2]

**(ii)** Determine the time taken for the discharge current to reduce to 15% of the initial discharge current.

time = ..... s [2]

[Total: 9]

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